

H2 Prelim 2 Paper 3

- 1 Phosphorus forms three chlorides, P_2Cl_2 , PCl_3 and PCl_5 , of which the latter two are the most common. Both PCl_3 and PCl_5 are valuable chemical intermediates in many processes.

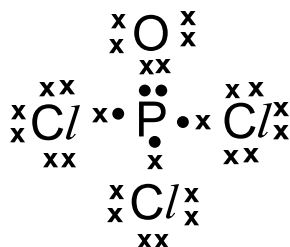
- (a) By considering bonding and structure, explain why PCl_3 is a gas while PCl_5 is a solid at room temperature and pressure. [2]

Both PCl_3 and PCl_5 are covalent simple molecules.

PCl_5 has a larger and more polarisable electron cloud than PCl_3 , resulting in stronger instantaneous dipole-induced dipole (id-id) interactions between PCl_5 molecules. More energy is required to overcome stronger id-id between PCl_5 molecules than that between PCl_3 molecules.

- (b) (i) Phosphorus oxychloride, $POCl_3$, is an intermediate compound formed during the reaction between PCl_5 and water.

Draw a dot-and-cross diagram to show the bonding in $POCl_3$ and state the shape of the molecule. [2]



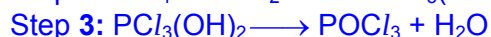
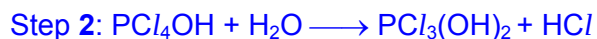
Shape : tetrahedral

- (ii) Write the equation of reaction of PCl_5 with water to form phosphorus oxychloride. [1]

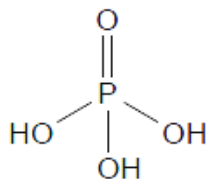


- (iii) Reactions of covalent chlorides with water can be rationalised as step-wise replacement of $-Cl$ with $-OH$.

There are 3 steps in the mechanism for the formation of phosphorus oxychloride from phosphorus pentachloride. Given step 1, complete the mechanism by writing the equations for steps 2 and 3.



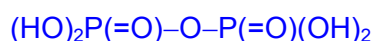
- (c) PCl_5 also reacts with water to form phosphoric acid, H_3PO_4 . The structure of phosphoric acid is shown below.



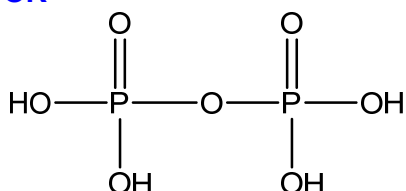
- (i) Phosphoric acid may dimerise to produce diphosphoric acid, $\text{H}_4\text{P}_2\text{O}_7$ and water. The reaction involves the condensation of an $-\text{OH}$ from each H_3PO_4 molecule to create an oxygen bridge between the two phosphoric acid units.

Draw the structure of diphosphoric acid.

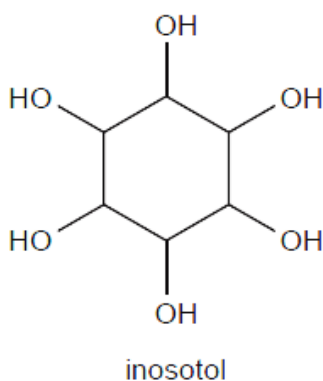
[1]



OR



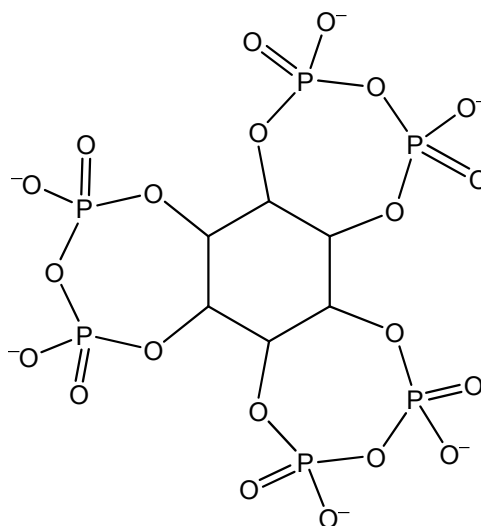
- (ii) Recent research from a group led by Nobel-prize winning chemist Jean-Marie Lehn has shown that a compound containing three cyclic diphosphates can improve the body's capacity for exercise. The compound is a derivative of inositol which is shown below.



In the new compound each pair of adjacent oxygen atoms from the inositol molecule is part of a cyclic diphosphate ester. The overall charge of the inositol-cyclic-phosphate species is -6 .

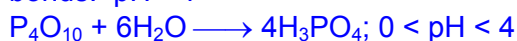
Suggest a structure for the compound.

[1]

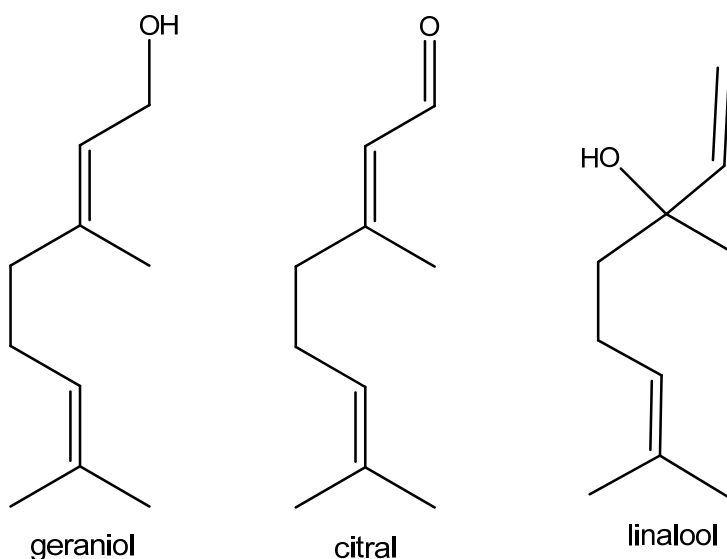


- (d) Describe the reactions, if any, of the oxides P_4O_{10} and SiO_2 with water. Include the approximate pH value of any resulting solutions, and write equations for any reactions that occur. [3]

SiO_2 does not dissolve in water due to network of numerous strong covalent bonds. pH = 7



- (e) Geraniol, citral and linalool are important flavours in citrus fruits. These compounds are often used in the food and perfume industries.



Suggest **two** simple chemical tests that could be used to distinguish the three compounds from each other. You should state what you would observe for **each** compound in **each** test. [4]

Test 1: Add Tollens' reagent to all compounds and warm.

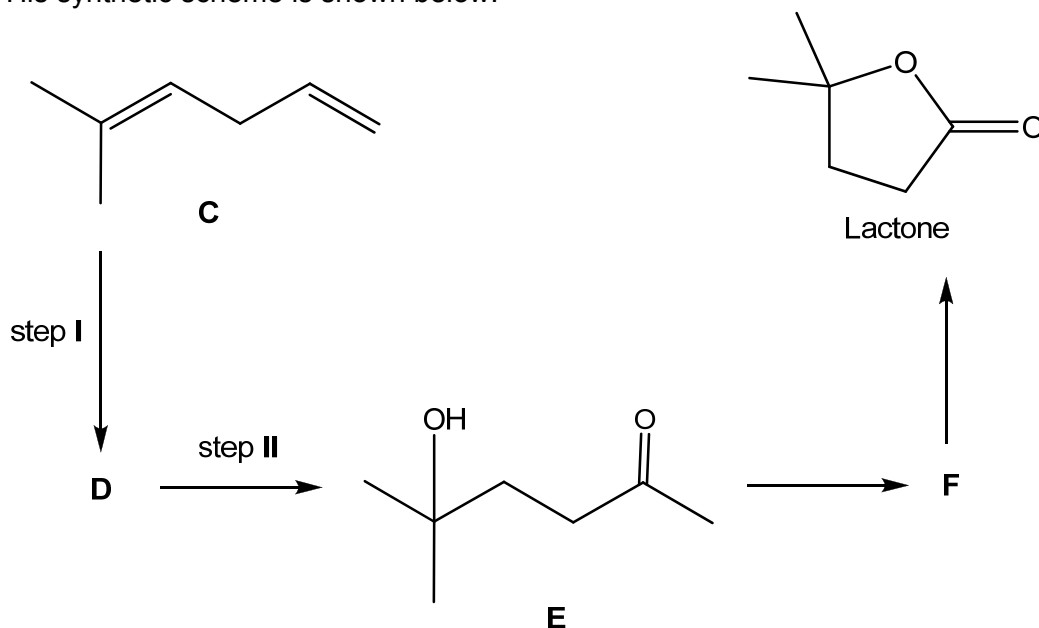
Observations: Only citral gives silver mirror but not linalool and geraniol /
Only citral gives silver mirror.

Test 2: Add $\text{K}_2\text{Cr}_2\text{O}_7$ (aq) and H_2SO_4 (aq) to all compounds and heat.

Observations: Both geraniol and citral turns orange solution green but not linalool.

Test	Geraniol	Citral	Linalool	Positive observation
Na(s), room temperature	✓		✓	Colourless gas that extinguishes lighted splint with 'pop' sound.
PCl ₅ , room temperature	✓		✓	White fumes of HCl
Fehling's reagent, warm		✓		Reddish brown ppt
2,4-dinitrophenylhydrazine		✓		Orange ppt
** Note that acidified KMnO ₄ , heat is not acceptable.				

- (f) A laboratory scientist attempted to synthesise a lactone starting from hydrocarbon **C**. His synthetic scheme is shown below.



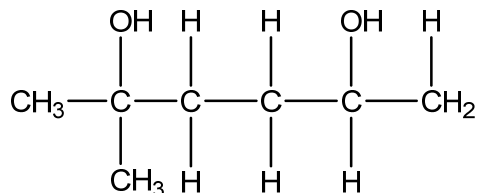
- (i) Suggest the reagents and conditions required for steps I and II. [2]

Step I: cold concentrated H₂SO₄, followed by H₂O, warm

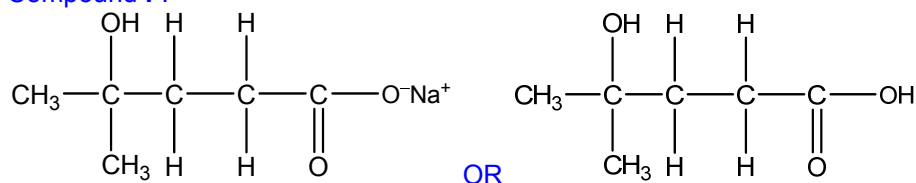
Step II: KMnO₄, H₂SO₄(aq), heat

- (ii) Give the structural formulae of compounds **D** and **F**. [2]

Compound **D**:



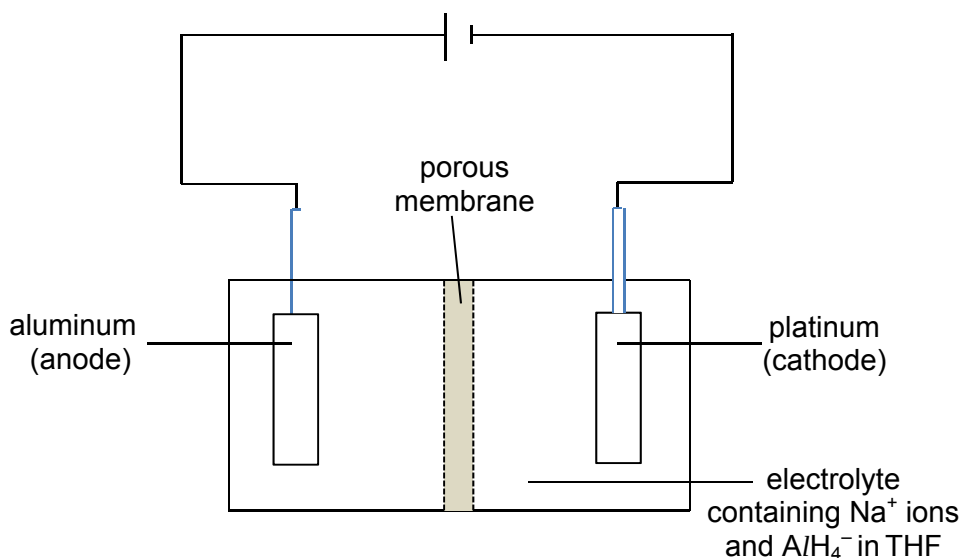
Compound **F**:



[Total: 20]

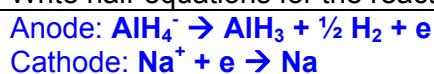
2	Aluminum is a silvery-white, soft metal which is the third most abundant element in the Earth's crust. Aluminum has a relatively low density, and is resistant to corrosion. Compounds of aluminum have also been used commercially for many purposes.		
	(a)	$\text{Al}(\text{OH})_3$ is a sparingly soluble aluminum compound with a K_{sp} value of 3×10^{-33} that can be used as a feedstock for manufacturing other aluminum compounds, such as aluminium chloride, zeolites, sodium aluminate.	
		(i)	Write an expression for the K_{sp} of $\text{Al}(\text{OH})_3$, stating the units.
			$K_{\text{sp}} = [\text{Al}^{3+}][\text{OH}^-]^3$ Unit: $\text{mol}^4 \text{dm}^{-12}$
		(ii)	Use the data above to determine the solubility of $\text{Al}(\text{OH})_3$ in water.
			$\text{Al}(\text{OH})_3(\text{s}) \rightleftharpoons \text{Al}^{3+}(\text{aq}) + 3\text{OH}^-(\text{aq})$ $x \qquad \qquad \qquad 3x$ Let the solubility of $\text{Al}(\text{OH})_3$ be $x \text{ mol dm}^{-3}$ $K_{\text{sp}} = [\text{Al}^{3+}][\text{OH}^-]^3$ $3 \times 10^{-33} = (x) \times (3x)^3$ $27x^4 = 3 \times 10^{-33}$ $x = 3.25 \times 10^{-9} \text{ mol dm}^{-3}$ Solubility = $3.25 \times 10^{-9} \text{ mol dm}^{-3}$
		(iii)	Predict, with a reason, how the solubility of $\text{Al}(\text{OH})_3$ will change when it is dissolved in 0.10 mol dm^{-3} of $\text{Al}(\text{NO}_3)_3$.
			$\text{Al}(\text{OH})_3(\text{s}) \rightleftharpoons \text{Al}^{3+}(\text{aq}) + 3\text{OH}^-(\text{aq})$ - The solubility of $\text{Al}(\text{OH})_3$ will <u>decrease</u> (due to common ion effect) - The <u>high $[\text{Al}^{3+}]$ from $\text{Al}(\text{NO}_3)_3$</u> will shift the <u>position of the above equilibrium to the left.</u>
		(iv)	$\text{Al}(\text{OH})_3$ is amphoteric in nature. Write balanced equations to illustrate this.
			It acts as a base when reacting with acid and acts as a base when reacting with acid $\text{Al}(\text{OH})_3(\text{s}) + \text{OH}^-(\text{aq}) \rightarrow [\text{Al}(\text{OH})_4]^-$ $\text{Al}(\text{OH})_3(\text{s}) + 3\text{H}^+(\text{aq}) \rightarrow \text{Al}^{3+} + 3\text{H}_2\text{O}$

- (b) Alane, AlH_3 , can also be produced under chemically inert environment from the electrolysis of a solution containing Na^+ ions and AlH_4^- ions in an organic solvent such as anhydrous tetrahydrofuran (THF). Aluminum is used as the anode, and platinum is used as the cathode. A simplified diagram of the electrochemical set-up is shown below.



Alane and hydrogen gas are produced at the anode, while sodium metal is collected at the cathode. THF is not involved in the reaction. A current of 2A is passed through the set-up for 40 minutes.

- (i) Write half-equations for the reactions at the anode and cathode respectively.



- (ii) A side-reaction occurred at the anode. It was observed that the anode became thinner over time. Given that alane is the only product in this side-reaction, suggest an ion-electron equation to account for this observation.



Note: $2\text{Al} + 3\text{H}_2 \rightarrow 2\text{AlH}_3$ is not acceptable as it is not an ion-electron equation.

- (iii) Based on the relevant equation in (i), calculate the amount of sodium collected from the experiment above.

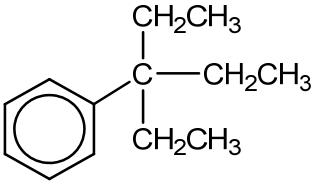
$$Q = It = n_e F$$

$$(2)(40 \times 60) = n_e (96500)$$

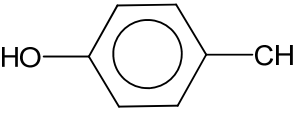
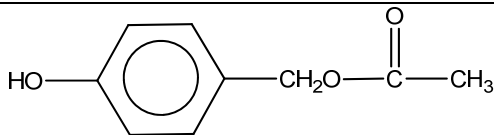
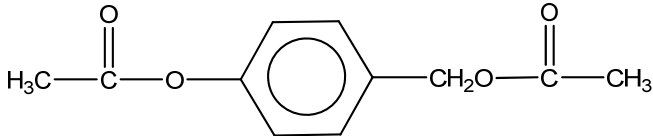
$$n_e = \frac{(2)(40 \times 60)}{96500} = 0.04974 \text{ mol}$$

Mole ratio of e^- : Na = 1:1
 Amount of sodium collected = 0.04974 mol

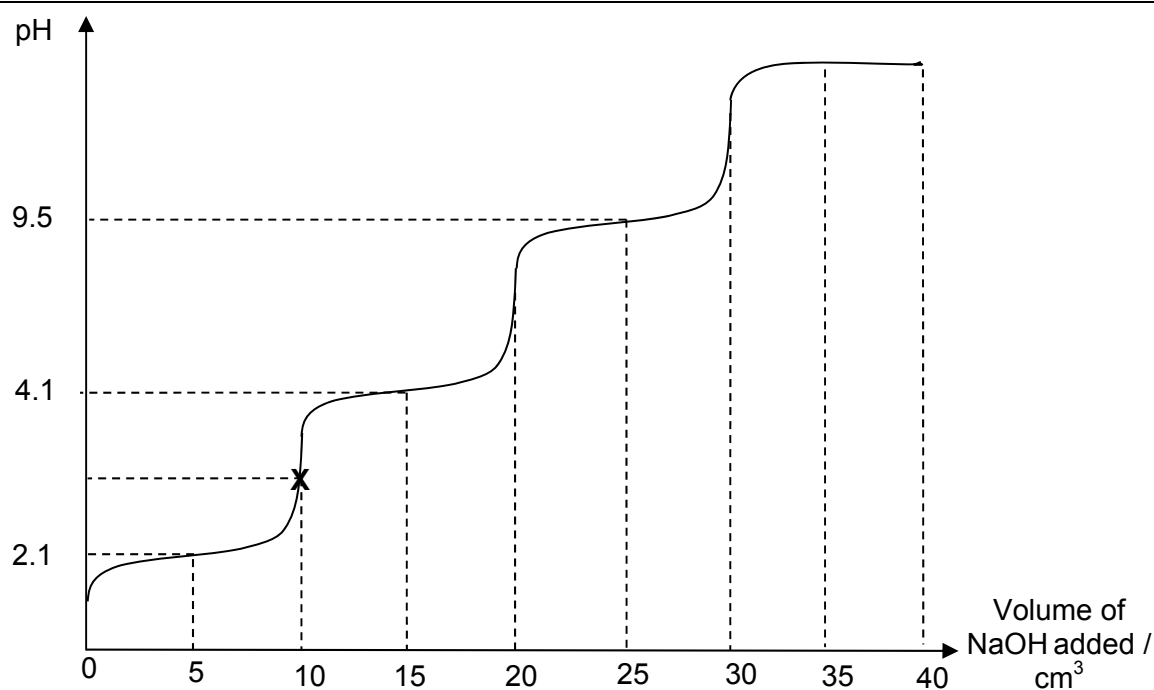
		(iv)	At room temperature and pressure, 473 cm ³ of hydrogen gas is formed at the anode. Based on your equations in (i) and (ii), calculate the amount of alane produced from the side-reaction.
			Amount of hydrogen gas formed = $\frac{473}{24000} = 0.01971 \text{ mol}$ Mole ratio of H₂: e = 1:2 Amount of electrons produced from reaction = $0.01971 \times 2 = 0.03942 \text{ mol}$ Hence, Amount of electrons produced from side-reaction = $0.04974 - 0.03942 = 0.01032 \text{ mol}$ Amount of alane produced from side-reaction = $0.01032 \div 3 = 0.003440 \text{ mol}$
		(c)	AlH₃ can be used as a reagent in organic reactions. Compound A can be synthesised using AlH₃ in the following reaction. $\text{NCCH}_2\text{CH}_2\text{CHO} \xrightarrow{\text{AlH}_3} \text{A} \quad (\text{C}_4\text{H}_{11}\text{NO})$ A shows negative result with Brady's reagent and reacts with sodium metal to produce a gas. Suggest the structure of A, and explain your answer.
			A shows negative result with Brady's reagent → A does not contain carbonyl group/ aldehyde and ketone group A reacts with sodium metal to produce a gas. → A reacts with Na via acid-metal reaction to form H₂ gas, → A contains alcohol group A contains 6 more H → A undergoes reduction with AlH₃, -CN is reduced to -CH₂NH₂. Therefore, structure of A: CH₂(OH)CH₂CH₂CH₂NH₂
		(d)	AlCl₃ can be used as a catalyst for the reaction of 2-chloro-3-methylbutane with benzene to form B. $ \begin{array}{c} \text{CH}_3 \text{ H} \\ \quad \\ \text{C} - \text{C} - \text{CH}_3 \\ \quad \\ \text{CH}_3 \text{ Cl} \end{array} \xrightarrow{\text{AlCl}_3} \begin{array}{c} \text{CH}_3 \text{ H} \\ \quad \\ \text{C} - \text{C} - \text{CH}_3 \\ \quad \\ \text{CH}_3 \text{ H} \end{array} $ B The reaction proceeds through the formation of a carbocation 1. As carbocation 1 is not very stable, a carbocation rearrangement occurs. In the rearrangement, a hydride shift is involved to form a more stable carbocation 2.
		(i)	Suggest the structures of both carbocations. Label your answers clearly, and explain why carbocation 2 is more stable than carbocation 1.
			$ \begin{array}{c} \text{CH}_3 \text{ H} \\ \quad \\ \text{H} - \text{C} - \text{C} - \text{CH}_3 \\ \quad \\ \text{CH}_3 \quad + \end{array} $ carbocation 1 $ \begin{array}{c} \text{CH}_3 \\ \\ +\text{C} - \text{CH}_2\text{CH}_3 \\ \\ \text{CH}_3 \end{array} $ carbocation 2

			Carbocation 2 contains 1 more electron-donating alkyl group attached to the positive carbon atom than that of carbocation 1. OR Carbocation 2 is a tertiary carbocation whereas carbocation 1 is a secondary carbocation. Hence, the positive charge on the carbon atom is more dispersed in carbocation 2.
		(ii)	Suggest the structure of product formed if compound E is used instead of 2-chloro-3-methylbutane for the above reaction. $ \begin{array}{c} \text{CH}_2\text{CH}_3 \\ \\ \text{H}-\text{C}-\text{CH}(\text{Cl})\text{CH}_3 \\ \\ \text{CH}_2\text{CH}_3 \\ \text{E} \end{array} $
			
			[Total: 20]

3	(a)	The graph below shows the enthalpy change of vaporisation of CH_3COOH , CH_3COSH and CH_3COSeH .								
		Enthalpy change of vaporisation/ kJ mol^{-1} <table border="1"> <thead> <tr> <th>Compound</th> <th>Enthalpy change of vaporisation / kJ mol^{-1}</th> </tr> </thead> <tbody> <tr> <td>CH_3COOH</td> <td>~27</td> </tr> <tr> <td>CH_3COSH</td> <td>~12</td> </tr> <tr> <td>CH_3COSeH</td> <td>~22</td> </tr> </tbody> </table>	Compound	Enthalpy change of vaporisation / kJ mol^{-1}	CH_3COOH	~27	CH_3COSH	~12	CH_3COSeH	~22
Compound	Enthalpy change of vaporisation / kJ mol^{-1}									
CH_3COOH	~27									
CH_3COSH	~12									
CH_3COSeH	~22									
	(i)	Explain the trend above. [3]								
		All three molecules have simple molecular structure. For CH_3COOH, there exist hydrogen bonds between molecules. More energy is needed to overcome the stronger hydrogen bonds thus it has a higher enthalpy of vaporization. CH_3COSeH has a larger electron cloud than CH_3COSH and is more easily polarized thus the strength of van der Waals' forces are stronger between CH_3COSeH molecules compared to between CH_3COSH. More energy is needed to overcome the intermolecular forces and CH_3COSeH has a higher enthalpy of vaporization.								
	(ii)	Ethanoic acid is found to have an apparent molar mass of 120 g mol^{-1} when it is heated to just above its boiling point. Draw a labelled diagram to illustrate this phenomenon. [1]								
		hydrogen bonding show lone pair of electrons on O atom show δ^- and δ^+ as shown in answer above for drawing dashed line (and label hydrogen bonding)								

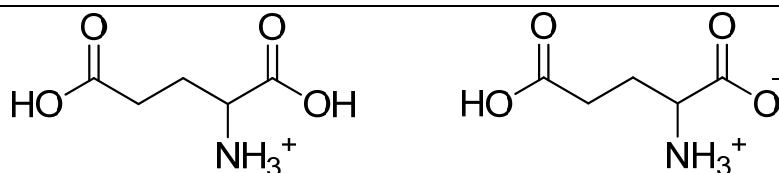
		(iii)	By making use of the above graph and your answer to (i), deduce the enthalpy change of hydrogen bond formation.	
				[2]
			Increase in enthalpy from CH_3COSH to $\text{CH}_3\text{COSeH} = 21.5 - 12 = 9.5 \text{ kJ mol}^{-1}$ Based on the graph, if the relationship is linear, CH_3COOH should have an enthalpy of vaporization of $(12 - 9.5) = 2.5 \text{ kJ mol}^{-1}$ (or read off from the graph) Enthalpy of formation of H-bonding $= -(26.5 - 2.5) = -24.0 \text{ kJ mol}^{-1}$	
		(iv)	Hence, or otherwise, determine the enthalpy change of dimerisation of ethanoic acid.	[1]
			2 hydrogen bonds formed upon dimerisation Enthalpy change of dimerisation $= 2(-24) = -48.0 \text{ kJ mol}^{-1}$ (must have correct sign)	
		(v)	Given that the entropy change of dimerisation is $-130 \text{ kJ mol}^{-1} \text{ K}^{-1}$, explain why dimerisation is spontaneous at low temperature under the same pressure.	[2]
			$\Delta G^\theta = \Delta H^\theta - T\Delta S^\theta$ $\Delta H^\theta < 0$, $\Delta S^\theta < 0$ - Since ΔS^θ is negative, $(-T\Delta S^\theta)$ is positive - At <u>low</u> temperatures, the <u>magnitude of ΔH^θ is larger than $(-T\Delta S^\theta)$</u>. ΔG^θ is <u>negative</u>. - Reaction is spontaneous only at <u>low</u> temperature.	
	(b)		Ethanoic acid and its derivative can react with 4-(hydroxymethyl)phenol to form two different esters under suitable conditions. $\text{Ester G} \xleftarrow[\text{conc H}_2\text{SO}_4]{\text{CH}_3\text{COOH}}$  $\xrightarrow[\text{NaOH (aq)}]{\text{CH}_3\text{COC/}}$ Ester H 4-(hydroxymethyl)phenol	
		(i)	Draw the structural formulae of ester G and H .	[2]
			G:  H: 	
		(ii)	Briefly explain why an alkaline medium is necessary for the formation of ester H .	[1]

			NaOH will react with the phenol to produce phenoxide ion which is a stronger nucleophile than phenol .												
	(c)	Collagen, a component of bone and connective tissue, is the most abundant protein in vertebrates. It exists as water-insoluble fibres of great strength. Five of the most common amino acids in the collagen obtained from human skin are listed below. <table><tr><th>Amino acid</th><th>Formula of side chain (R in RCH(NH₂)COOH)</th></tr><tr><td>glycine (gly)</td><td>–H</td></tr><tr><td>alanine (ala)</td><td>–CH₃</td></tr><tr><td>glutamic acid (glu)</td><td>–CH₂CH₂COOH</td></tr><tr><td>serine (ser)</td><td>–CH₂OH</td></tr><tr><td>lysine (lys)</td><td>–(CH₂)₄NH₂</td></tr></table>		Amino acid	Formula of side chain (R in RCH(NH ₂)COOH)	glycine (gly)	–H	alanine (ala)	–CH ₃	glutamic acid (glu)	–CH ₂ CH ₂ COOH	serine (ser)	–CH ₂ OH	lysine (lys)	–(CH ₂) ₄ NH ₂
Amino acid	Formula of side chain (R in RCH(NH ₂)COOH)														
glycine (gly)	–H														
alanine (ala)	–CH ₃														
glutamic acid (glu)	–CH ₂ CH ₂ COOH														
serine (ser)	–CH ₂ OH														
lysine (lys)	–(CH ₂) ₄ NH ₂														
	(i)	There are three pK_a values associated with glutamic acid: 2.1, 4.1 and 9.5 Sketch the pH-volume curve obtained when 10 cm³ of 0.1 mol dm^{–3} protonated glutamic acid is titrated against 0.1 mol dm^{–3} NaOH(aq) until 40 cm³ of NaOH(aq) is added.													
			[3]												



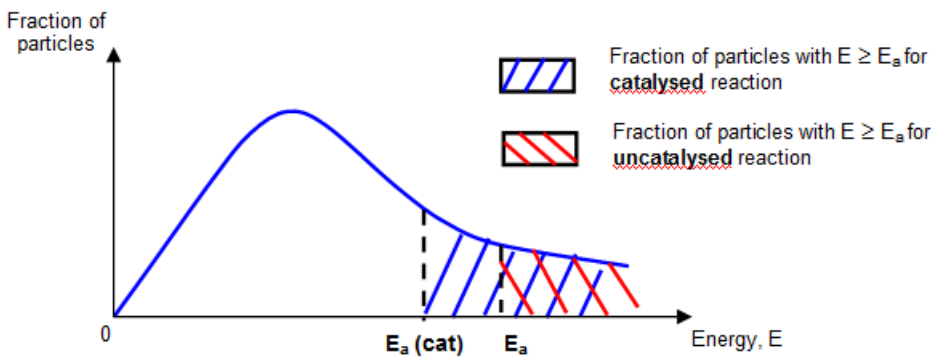
correct axes
 graph should end at least at 40cm³ mark
 correct sketch of graph with 3 end points
 each end point correspond to 10, 20 and 30 cm³
 all three MBC indicated
 all three MBC corresponds to 5, 15 and 25 cm³

(ii) Draw the two major forms of glutamic acid present at pH 2.1. [2]

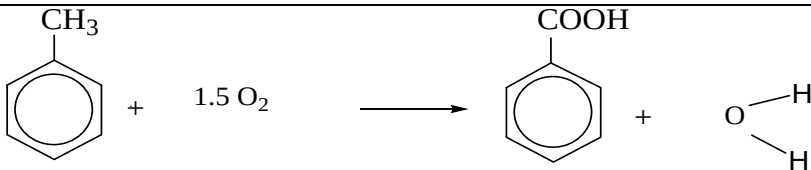
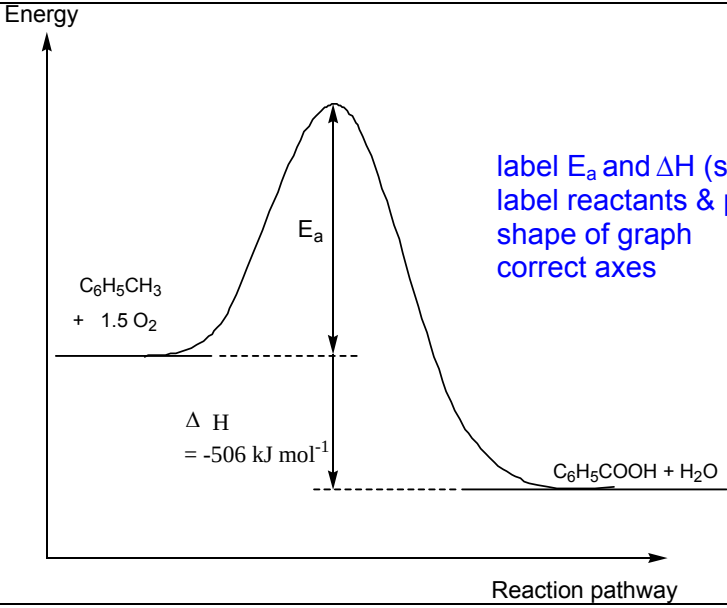


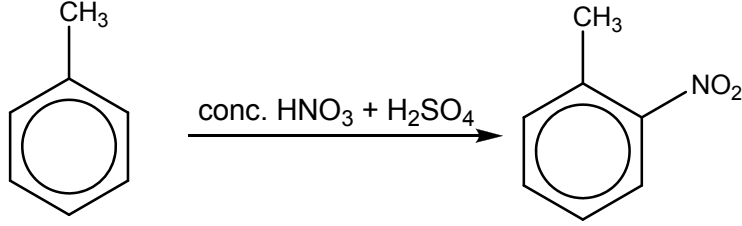
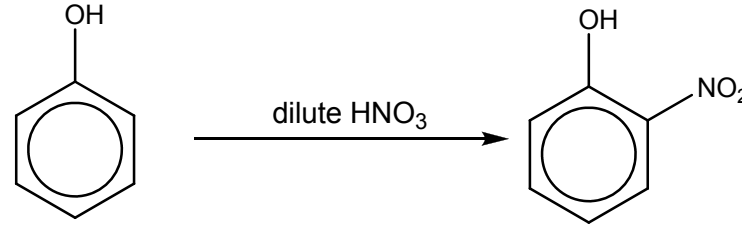
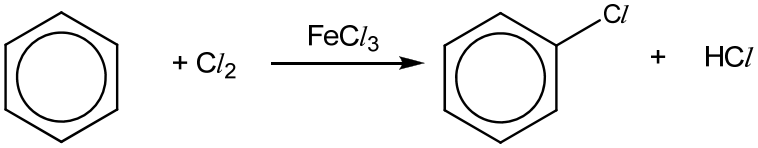
(d) (i) Human haemoglobin undergoes denaturation in a strongly basic medium. With respect to the globular protein subunit, explain why denaturation takes place. [2]

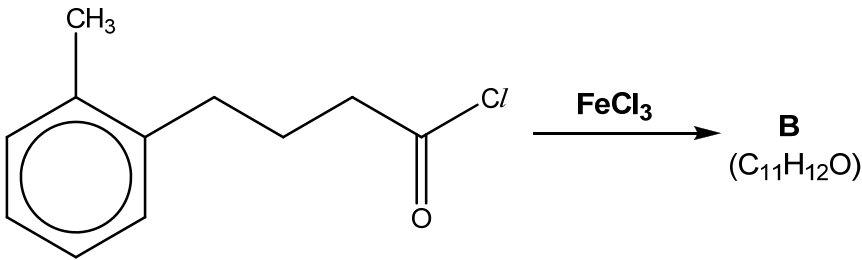
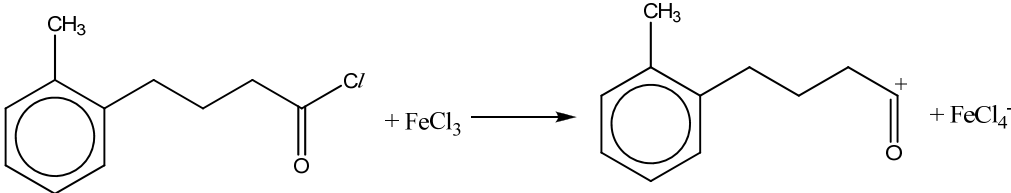
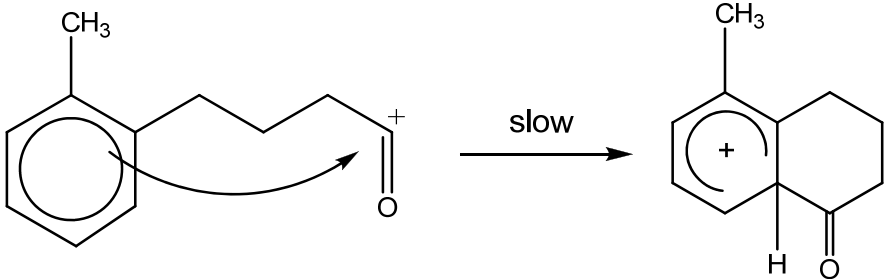
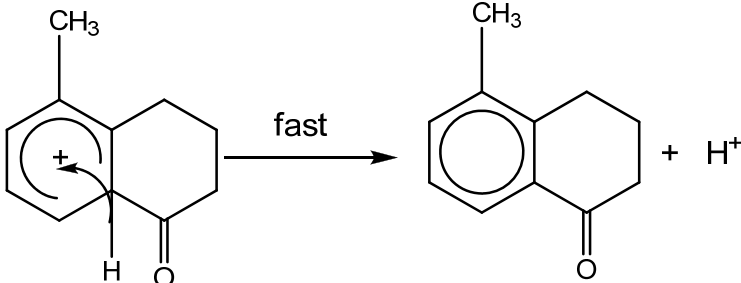
The acidic R groups, COOH/NH₃⁺ are deprotonated, which affect the ionic interactions/hydrogen bonding causing the unfolding of the protein chain.

	(ii)	Hemochromatosis is an autosomal recessive genetic disease that results in deregulated iron absorption in specific organs such as the liver and heart. Liver enzymes acts as a catalyst and an increase in amount of the enzymes will aggravate the condition of a patient diagnosed with hemochromatosis. Sketch the Boltzmann distribution curve to show how the presence of the liver enzyme aggravates the condition.
		[1]
		 Origin + both axes + graph start at origin Correct legends and shading must tally with graph Correct shape of graph with labels of $E_a(\text{cat})$ and E_a Labelling of $E_a(\text{cat})$ must be lower than E_a
		[Total: 20]

4	(a)	Methylbenzene is a colourless aromatic hydrocarbon that is obtained from petroleum by distillation. It is used as a solvent and as a starting material for the synthesis of many organic compounds, including dyes and explosives. It can be oxidised to benzoic acid using oxygen over a suitable catalyst according to the following equation. $\text{C}_6\text{H}_5\text{CH}_3(\text{g}) + 1.5\text{O}_2(\text{g}) \longrightarrow \text{C}_6\text{H}_5\text{COOH}(\text{g}) + \text{H}_2\text{O}(\text{g})$
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		(i)	Use bond energy values from the Data Booklet to calculate the enthalpy change for the reaction.																												
			 <table border="1" style="margin: 10px auto; border-collapse: collapse; text-align: center;"><thead><tr><th style="color: blue;">Bonds broken</th><th style="color: blue;">No. of moles</th><th style="color: blue;">Bond energy / kJ mol⁻¹</th><th></th><th style="color: blue;">Bonds formed</th><th style="color: blue;">No. of moles</th><th style="color: blue;">Bond energy / kJ mol⁻¹</th></tr></thead><tbody><tr><td style="color: blue;">O=O</td><td style="color: blue;">1.5</td><td style="color: blue;">496x1.5</td><td></td><td style="color: blue;">O-H</td><td style="color: blue;">3</td><td style="color: blue;">460x3</td></tr><tr><td style="color: blue;">C-H</td><td style="color: blue;">3</td><td style="color: blue;">410x3</td><td></td><td style="color: blue;">C-O</td><td style="color: blue;">1</td><td style="color: blue;">360</td></tr><tr><td></td><td></td><td></td><td></td><td style="color: blue;">C=O</td><td style="color: blue;">1</td><td style="color: blue;">740</td></tr></tbody></table> $\Delta H = \Sigma \text{ B.E. (bonds broken)} - \Sigma \text{ B.E. (bonds formed)}$ $= 1974 - 2480$ $= -506 \text{ kJ mol}^{-1}$	Bonds broken	No. of moles	Bond energy / kJ mol ⁻¹		Bonds formed	No. of moles	Bond energy / kJ mol ⁻¹	O=O	1.5	496x1.5		O-H	3	460x3	C-H	3	410x3		C-O	1	360					C=O	1	740
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		(ii)	Using your answer from (a)(i) , construct a reaction pathway diagram for this reaction.																												
			 label E_a and ΔH (single head arrows) label reactants & products shape of graph correct axes																												
			[4]																												
		(b)	Both methylbenzene and phenol can react with nitric acid, but under different conditions.																												

		 
	(i)	Explain why different conditions are needed for these two reactions.
		With phenol, catalyst H_2SO_4 is not required because the delocalization of the lone pair of electrons on the oxygen atom into benzene ring causes the ring to be more electron rich. Hence, phenol is more susceptible to electrophilic attacks and thus there was no longer the need for the strong electrophile NO_2^+, created by the reaction between HNO_3 and H_2SO_4.
	(ii)	Suggest and explain how the acidity of 2-nitrophenol might compare with that of phenol.
		• Nitro is an electron withdrawing group which • further disperses the negative charge on O atom of 2-nitrophenoxide ion • hence, 2-nitrophenoxide is more stable than phenoxide ion • 2-nitrophenol is more acidic
		[4]
	(c)	Iron (III) chloride is used as a catalyst in electrophilic substitution reactions. The chlorination of benzene is represented by the following overall equation.  The reaction occurs in several steps. • The first step is the reaction between Cl_2 and FeCl_3. • The benzene ring is then attacked by the Cl^+ cation in the second step. FeCl_3 reacts in a similar way with acyl chlorides, producing a carbocation that can then attack a benzene ring.
		Describe the mechanism for the following reaction:

		 B (C₁₁H₁₂O)
		Step 1: Generation of electrophile  Step 2: Electrophilic attack on benzene ring  Step 3: Loss of proton  Step 4: Regeneration of catalyst $\text{H}^+ + [\text{FeCl}_4]^- \longrightarrow \text{FeCl}_3 + \text{HCl}$
		[2]
	(d)	A 0.150 mol dm ⁻³ solution of iron(III) chloride was found to have a pH of 1.52.
	(i)	Write an expression for the acid dissociation constant K _a of [Fe(H ₂ O) ₆] ³⁺ .
		$K_a = \frac{[\text{Fe}(\text{H}_2\text{O})_5\text{OH}]^{2+} [\text{H}^+]}{[\text{Fe}(\text{H}_2\text{O})_6]^{3+}}$
	(ii)	Calculate the K _a of [Fe(H ₂ O) ₆] ³⁺ .
		$\text{pH} = -\log \sqrt{K_a [\text{acid}]}$

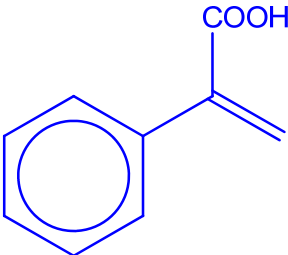
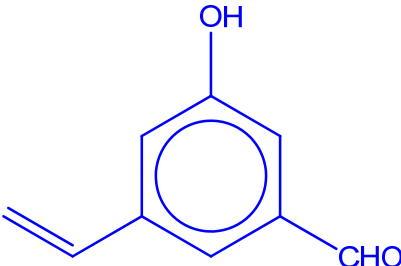
			$1.52 = -\log \sqrt{K_a(0.150)}$ $K_a = 6.08 \times 10^{-3} \text{ moldm}^{-3}$
		(iii)	Explain why the pH of a solution of iron(II) chloride is higher than that of a solution of iron (III) chloride of the same concentration.
			Fe^{2+} has a smaller charge density and thus less polarising than Fe^{3+} , and hence it hydrolyses in water to a smaller extent to give H^+ ions.
			[3]
	(e)	<i>Use of the Data Booklet is relevant to this question.</i> When aqueous iron(II) chloride is boiled with an excess of potassium cyanide, a yellow solution D is obtained. Acidification of D, followed by the passage of chlorine gas through the solution, gives an orange solution E. Iron(II) chloride is found to catalyse the reaction between I^- and $\text{S}_2\text{O}_8^{2-}$ whereas E does not affect the rate of this reaction. Solution E can also be formed by adding aqueous potassium cyanide to brown precipitate B. Precipitate B is formed by adding excess NH_3 to aqueous iron(II) chloride and letting it stand in air.	
		(i)	Explain why a solution of iron(II) chloride appear green.
			$\text{FeCl}_2(\text{aq})$ is coloured due to: ✓ presence of <u>incompletely filled 3d orbitals</u> in the <u>Fe^{2+} ions.</u> ✓ In the <u>presence of ligands</u>, <u>3d orbitals split into two groups with small energy gap</u> ✓ Some of the <u>light energy is used to promote an electron from a 3d orbital of lower energy into the partially filled 3d orbital of higher energy.</u> ✓ Colour seen is the <u>complement of the RED colour absorbed in the visible region</u> of the spectrum
		(ii)	Identify solution D and suggest the type of reaction that has occurred for the formation of D from iron(II) chloride solution.
			D: $[\text{Fe}(\text{CN})_6]^{4-}$ Type of reaction: Ligand exchange
		(iii)	Identify precipitate B .
			B: $\text{Fe}(\text{OH})_3$

		(iv)	Identify solution E .
			E: $[\text{Fe}(\text{CN})_6]^{3-}$
		(v)	Explain why E does not affect the rate of reaction between I^- and $\text{S}_2\text{O}_8^{2-}$.
			$[\text{Fe}(\text{CN})_6]^{3-}$ is unable to oxidise I^- to I_2 since the $E^\ominus[\text{Fe}(\text{CN})_6]^{3-}/[\text{Fe}(\text{CN})_6]^{4-}$ is less positive than +0.54V OR E_{cell} for reaction between $[\text{Fe}(\text{CN})_6]^{3-}$ and I^- is less than zero.
			[7]
			[Total: 20]

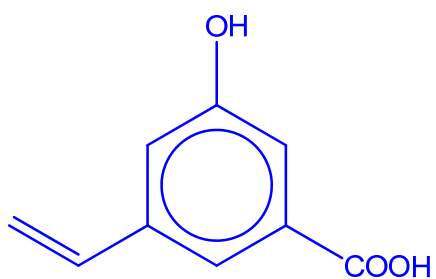
5	The halogens and their compounds are useful laboratory reagents.	
	(a)	Describe how the reactions of chloride and iodide ions with concentrated H_2SO_4 differ. You should include suitable observations and balanced equations for any reactions that occur.
		Reaction of Cl^- with conc. H_2SO_4: $\text{Cl}^- + \text{H}_2\text{SO}_4 \rightarrow \text{HCl} + \text{HSO}_4^-$ Observation: Steamy fumes of HCl (g) seen Reaction of I^- with conc. H_2SO_4:

		$\text{I}^- + \text{H}_2\text{SO}_4 \rightarrow \text{HI} + \text{HSO}_4^-$ $8\text{HI} + \text{H}_2\text{SO}_4 \rightarrow 4\text{I}_2 + \text{H}_2\text{S} + 4\text{H}_2\text{O}$ OR $2\text{HI} + \text{H}_2\text{SO}_4 \rightarrow \text{I}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$ OR $6\text{HI} + \text{H}_2\text{SO}_4 \rightarrow 3\text{I}_2 + \text{S} + 4\text{H}_2\text{O}$ Observation: Violet fumes of I_2 and some steamy white fumes of HI evolved. OR Pungent SO_2 evolved OR Pungent H_2S evolved (rotten eggs smell) OR Yellow solid of S is produced.	[4]
	(b)	Phenol and chlorobenzene are less reactive towards certain reagents than similar non-aromatic compounds. Thus hexan-1-ol can be converted into hexylamine by the following two reactions: $\begin{array}{ccccc} \text{CH}_3(\text{CH}_2)_5\text{OH} & \xrightarrow{\text{I}} & \text{CH}_3(\text{CH}_2)_5\text{Cl} & \xrightarrow{\text{II}} & \text{CH}_3(\text{CH}_2)_5\text{NH}_2 \\ \text{hexan-1-ol} & & \text{1-chlorohexane} & & \text{hexylamine} \end{array}$ whereas neither of the following two reactions takes place. $\text{C}_6\text{H}_5\text{OH} \not\rightarrow \text{C}_6\text{H}_5\text{Cl} \not\rightarrow \text{C}_6\text{H}_5\text{NH}_2$	
	(i)	Suggest reagents and conditions for reactions I and II.	[2]
		Reaction I: Reagents and conditions: Conc HCl , anhydrous ZnCl_2 catalyst, heat with reflux OR PCl_5 , room temperature OR PCl_3 , room temperature OR SOCl_2 , warm Reaction II: Reagents and conditions: Excess ethanolic NH_3 , heat in sealed tube	
	(ii)	What type of reaction is reaction II?	[1]
		Nucleophilic substitution	
	(iii)	Suggest a reason why chlorobenzene is much less reactive than 1-chlorohexane.	

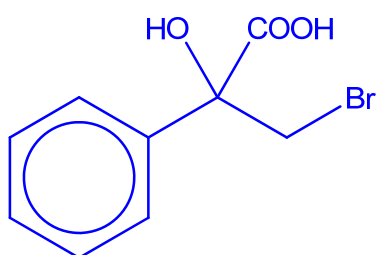
				[2]										
			<u>The p-orbital of the halogen atom overlaps with π-orbitals of the benzene ring to form a delocalised π electron cloud / results in delocalization of π electron.</u> The delocalisation of electrons <u>shortens and strengthens the carbon-chlorine bond, making it too strong to be broken easily</u> , therefore chlorobenzene is much less reactive than 1-chlorohexane.											
		(iv)	State and explain how the rate of reaction II changes when 1-chlorohexane is replaced by 1-iodohexane.	[2]										
			The <u>rate of the reaction will be faster</u> when it is replaced by 1-iodohexane. The <u>bond length of C–I bond is longer than C–Cl bond</u> and hence the <u>C–I bond is weaker</u> . <u>Less energy is required to break the C–I bond</u> , thus 1-iodohexane is more reactive.											
		(c)	The compounds P and Q are isomers with the molecular formula $C_9H_8O_2$. P liberates carbon dioxide with sodium hydrogencarbonate but Q does not. Q reacts with an ammoniacal solution of silver nitrate followed by acidification to give R and a grey precipitate. Upon oxidation with hot acidified potassium manganate(VII), 1 mole of P gives 2 moles of carbon dioxide and 1 mole of benzoic acid while 1 mole of Q gives 1 mole of carbon dioxide and 1 mole of a compound, $C_8H_6O_5$. P decolourises aqueous bromine in the dark to form S, $C_9H_9O_3Br$ while Q reacts with aqueous bromine to form a white precipitate T, $C_9H_6O_3Br_4$. When compound S is heated with acidified sodium dichromate (VII) solution, the solution remains orange. Deduce the structures of all the lettered compounds (P to T) and explain the chemistry of the reactions described.											
			<table><thead><tr><th>Observation</th><th>Deduction</th></tr></thead><tbody><tr><td>P and Q have molecular formula $C_9H_8O_2$.</td><td>Both P and Q contain a <u>benzene ring</u> due to the <u>C to H ratio ~ 1:1</u>.</td></tr><tr><td>P gives $CO_2(g)$ upon reaction with $NaHCO_3$ but not Q.</td><td>P undergoes <u>acid-base reaction</u> with $NaHCO_3$ but not Q. $\Rightarrow$ P is a <u>carboxylic acid</u> but Q is not a carboxylic acid.</td></tr><tr><td>Q reacts with ammoniacal solution of $AgNO_3$ to give R.</td><td>Q undergoes <u>oxidation</u> with Tollens' reagent to give a grey precipitate of Ag and R, a carboxylic acid after acidification. $\Rightarrow$ Q is an <u>aldehyde</u>.</td></tr><tr><td>Upon oxidation with hot acidified $KMnO_4$, 1 mole of P gives 2 moles of CO_2 and 1 mole of benzoic acid</td><td>Oxidation of side-chain occurs. $\Rightarrow$ P contains <u>a benzene ring with only 1 side chain</u>. Hence P can either be  $-CH=CH-COOH$ or  $-CH_2-COOH$ </td></tr></tbody></table>	Observation	Deduction	P and Q have molecular formula $C_9H_8O_2$.	Both P and Q contain a <u>benzene ring</u> due to the <u>C to H ratio ~ 1:1</u> .	P gives $CO_2(g)$ upon reaction with $NaHCO_3$ but not Q .	P undergoes <u>acid-base reaction</u> with $NaHCO_3$ but not Q . $\Rightarrow$ P is a <u>carboxylic acid</u> but Q is not a carboxylic acid.	Q reacts with ammoniacal solution of $AgNO_3$ to give R .	Q undergoes <u>oxidation</u> with Tollens' reagent to give a grey precipitate of Ag and R , a carboxylic acid after acidification. $\Rightarrow$ Q is an <u>aldehyde</u> .	Upon oxidation with hot acidified $KMnO_4$, 1 mole of P gives 2 moles of CO_2 and 1 mole of benzoic acid	Oxidation of side-chain occurs. $\Rightarrow$ P contains <u>a benzene ring with only 1 side chain</u> . Hence P can either be  $-CH=CH-COOH$ or  $-CH_2-COOH$	
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		Upon oxidation with hot acidified KMnO_4, 1 mole of Q gives 1 mole of CO_2 and 1 mole of a compound, $\text{C}_8\text{H}_6\text{O}_5$.	Oxidation of $\text{C}=\text{C}$ in Q occurs. $\Rightarrow$ Q contains a $\text{C}=\text{CH}_2$ structural unit which is oxidised to give $-\text{COOH}$ and CO_2. Oxidation of $-\text{CHO}$ in Q also occurs. $\Rightarrow$ The oxidation product contains 2 $-\text{COOH}$ groups. With a molecular formula of $\text{C}_8\text{H}_6\text{O}_5$, the oxidation product also has to contain a phenolic $-\text{OH}$ group. Hence the oxidation product contains a benzene ring with 3 substituents: 2 $-\text{COOH}$ groups and 1 $-\text{OH}$ group. $\Rightarrow$ Q contains <u>a benzene ring with 3 substituents: $-\text{CHO}$, $-\text{CH}=\text{CH}_2$ and $-\text{OH}$ groups.</u> Or Q contains a <u>terminal alkene</u> or $=\text{CH}_2$. Or <u>Oxidation of $\text{C}=\text{C}$ and $-\text{CHO}$/alkene and aldehyde.</u>	
		P decolourises $\text{Br}_2(\text{aq})$ in the dark to form S, $\text{C}_9\text{H}_9\text{O}_3\text{Br}$.	P undergoes <u>electrophilic addition</u>. $\Rightarrow$ P is an <u>alkene or contains a $\text{C}=\text{C}$ bond</u>. And S is a bromohydrin.	
		Q reacts with $\text{Br}_2(\text{aq})$ to form a white precipitate T, $\text{C}_9\text{H}_6\text{O}_3\text{Br}_4$.	Q also undergoes <u>electrophilic substitution</u>. $\Rightarrow$ Q is a <u>phenol</u> and its two substituents are at positions 3 & 5. Q undergoes electrophilic addition $\Rightarrow$ Q is an alkene or contains a $\text{C}=\text{C}$ bond.	
		When compound S is heated with acidified sodium dichromate (VII) solution, the solution remains orange.	Since no oxidation occurs, compound S must be a bromohydrin with a <u>tertiary alcohol</u>.	
		compound P: 	compound Q: 	

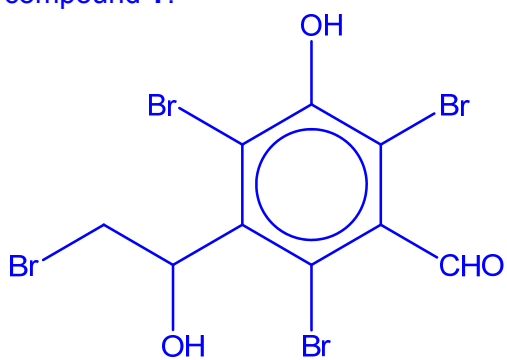
compound **R**:



compound **S**:



compound **T**:



[9]

[Total: 20]